GENERATION OF NORMAL DISTRIBUTION

PROBLEM

Flipper lengths of a certain kind of penguin are normally distributed with mean of 75 and a standard deviation of 10.

a. What is the probability of getting a score between 60 and 80?

b. What is the probability of getting a score above 90?

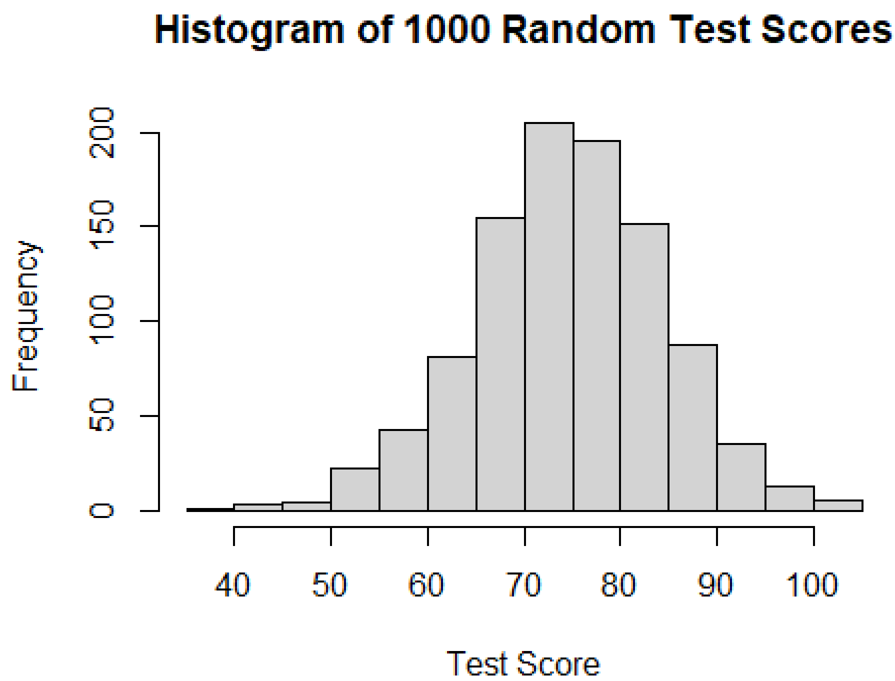
c. Simulate 1000 random selections from this population and plot the histogram of the results.

AIM

To calculate the probability of getting a score between 60 and 80, the probability of getting a score above 90, and to simulate 1000 random selections from this population and plot the histogram of the results.

CODE AND OUTPUT

```
#Set mean and standard deviation

mean=75
sd=10

#Calculate probability of score between 60 and 80

prob_between = pnorm(80, mean, sd) - pnorm(60, mean, sd)
cat("Probability of score between 60 and 80:", prob_between, "\\n")

## Probability of score between 60 and 80: 0.6246553 \n

## Probability of score between 60 and 80: 0.6246553

#Calculate probability of score above 90

prob_above_90 = 1 - pnorm(90, mean, sd)
cat("Probability of score above 90:", prob_above_90, "\\n")

## Probability of score above 90: 0.0668072 \n

## Probability of score above 90: 0.0668072

#Simulate 1000 random selections and plot histogram

scores = rnorm(1000, mean, sd)

hist(scores, main = "Histogram of 1000 Random Test Scores", xlab = "Test Score")
```



RESULT

Probability of randomly selecting a penguin having flipper length:

(a) between 60 and 80 = 0.6246553.

(b) getting a score above 90 = 0.0668072.

The histogram for normal distribution is been displayed.

LINEAR PROGRAMMING

PROBLEM

Imagine that you operate a furniture company, with the following three products

- Tables: Each table makes a profit of $500, costs 8.4 production hours to make, and $3m^3$ of storage to store
- Sofas: Each sofa makes a profit of $450, costs 6 production hours to make, and $6m^3$ of storage to store
- Beds: Each bed makes a profit of $580, costs 9.6 production hours to make, and $8m^3$ of storage to store

The profit earned by selling each type of funiture is listed above. However, each piece of furniture requires some factory-time to make and requires warehouse space to store.

Each week you have a budget of only 72 production hours at your factory. Additionally, you have enough warehouse storage to store only $40m^3$ (cubic-meters).

Furthermore, from past trends, you can only sell a maximum of 8 tables a week.

How many of each type of furniture should you make for next week to maximise profit?

AIM

To find the maximize of the profit using linear programming.

DECISION VARIABLES, OBJECTIVE AND CONSTRAINTS

First, we define our decision variables and objective function:

X1 = number of tables made

X2 = number of sofas made

X3 = number of beds made

Maximize Profit = 500X1+450X2+580X3

Next, we define our four constraints:

Production Hour Budget Constraints:

8.4X1+6X2+9.6X3≤72

Storage Space Constraints:

3X1+6X2+8X3≤40

Demand Constraints:

X1+X2+X3≤8

Non-negativityConstraints:

X1≥0; X2≥0, X3≥0

CODE AND OUTPUT

```
library(lpSolve)
objective.fn = c(500, 450, 580)
const.mat = matrix(c(8.4, 6, 9.6, 3, 6, 8, 1, 0, 0), ncol = 3, byrow = TRUE)
const.dir = c("<=", "<=", "<=")
const.rhs = c(72, 40, 8)
#solving model
lp.solution = lp("max", objective.fn, const.mat, const.dir, const.rhs, compute.sens = TRUE)
lp.solution

## Success: the objective function is 4629.63

lp.solution$solution

## [1] 5.925926  3.703704  0.000000
```

RESULT

Thus, the optimal solution is: X1=5.93, X2=3.70, X3=0

Hence, the optimal solution, given the constraints, is to produce 5.93 tables, 3.70 sofas, and 0 beds.

This will yield a maximum profit of $4629.63